Discounts and Premiums

Péter Juhász, PhD, CFA

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Why do we need D&Ps?

To include effects into our valuation that the given method has not yet (fully) considered.

The use and extent of discounts and premiums vary across valuation methods even within the same valuation task.

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Inclusion of key effects

Effects	DCF	Multiples	Asset-based
Business outlook	CF plan	Industry average	–
Market judgement	Discount rate	Market ratios	Market prices
Marketability, keyperson	**Premiums and discounts**		Different!
Ownership structure			None!
Off-balance sheet items, others			Should be already included

Each approach calls for a different correction.

DCF: Theoretical intrinsic value under perfect conditions (liquid markets, no information asymmetry)

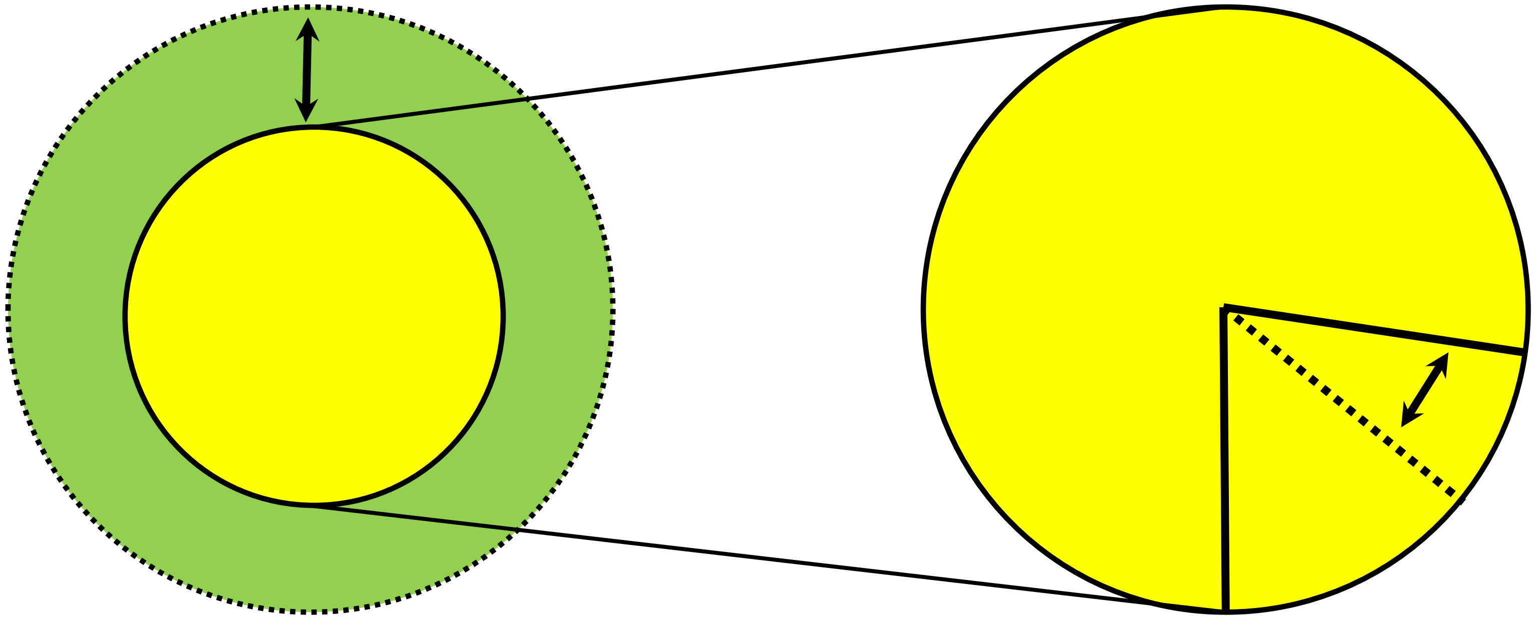
Market multiples: Average value of similar firms (typical, listed, liquid firm)

Comparable transactions: We are exactly like the average of the peers (same country, markets, size, liquidity)

Corrections should only hit the value of operation.

Corrections for equity value

Corrections for a given stake



D&Ps for total equity

Liquidity („illiquidity”) discount (Several meanings: assets, size, financial power, chance of going public.)

Key person discount

Complex portfolio without synergy

Environmental, legal, and dependent liabilities

Long term liabilities (contracts, warranties, guaranties, warrants)

D&Ps for an equity stake

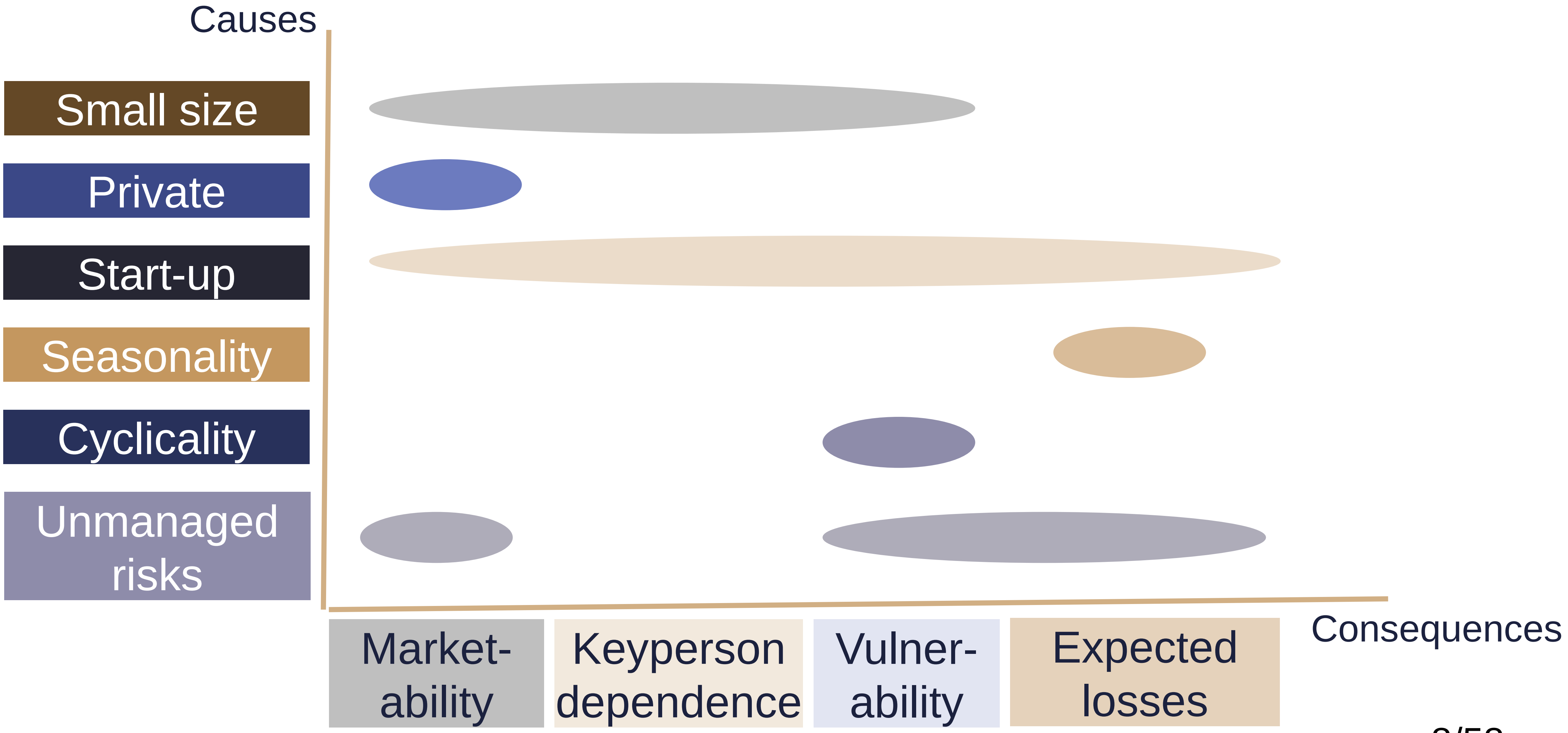
Liquidity discount (no buyer e.g. due to the ownership structure)

Control rights (minority, blocking minority, majority, strategic majority)

Additional voting right (preferred shares)

Big package discount (if sold current price would fall)

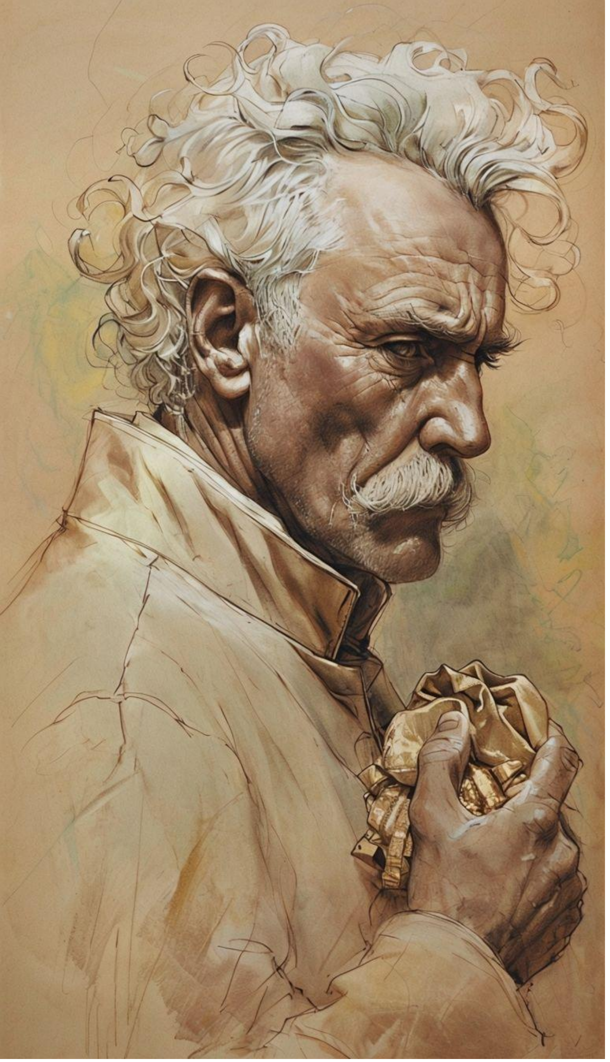
Do not mix the dimensions



The valuation process – Stakes and shares

1. Determining the value of operation (V_{op})
2. Valuing non-operative assets (NOA)
3. Calculating the total firm value ($V=V_{op}+NOA$)
4. Valuing the foreign capital (D)
5. Determining the market value of equity ($E=V-D$)
6. **Finding the price of a share or stake**

The two faces of control premium



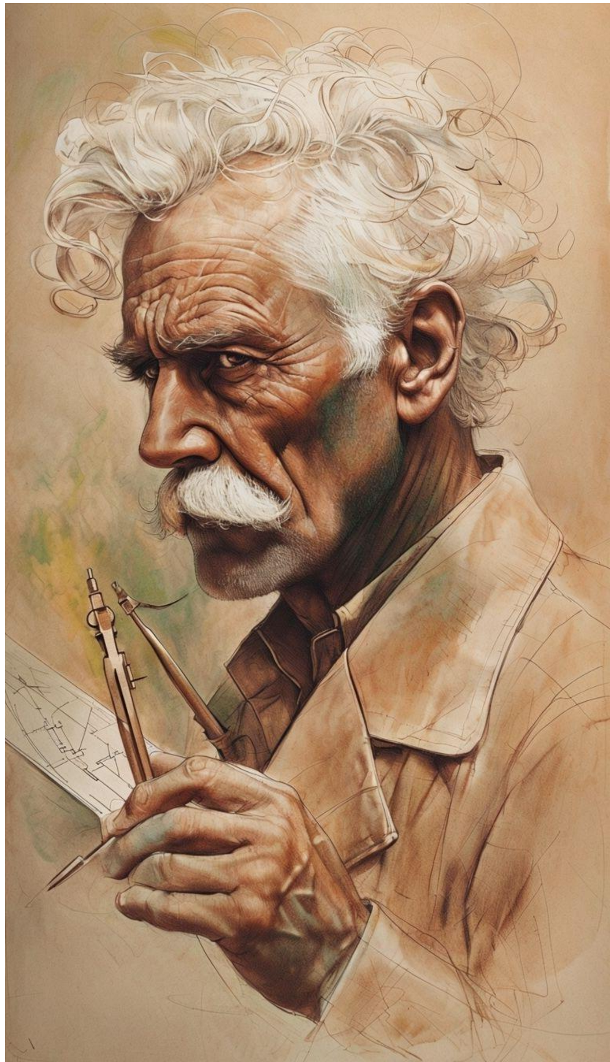
I. Rule

Divert CF

II. Change

Improve CF

The control premium we mentioned at the M&A class



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I. Valuing equity stakes – no ownership change

The value of a firm is 1200, its debt amounts to 200 and there are 2000 shares issues. What is the market price of a single share?

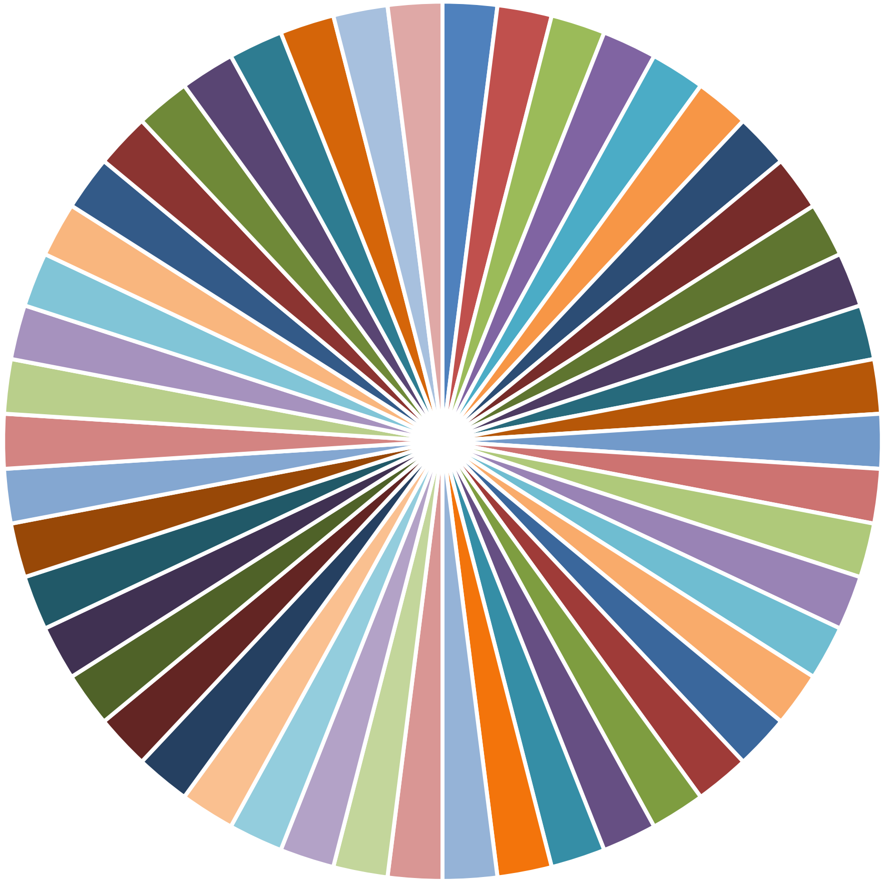
MV(D+E)-MV(D)= MV(E)

1200-200=1000

? **P=1000/2000=0.5** **?**

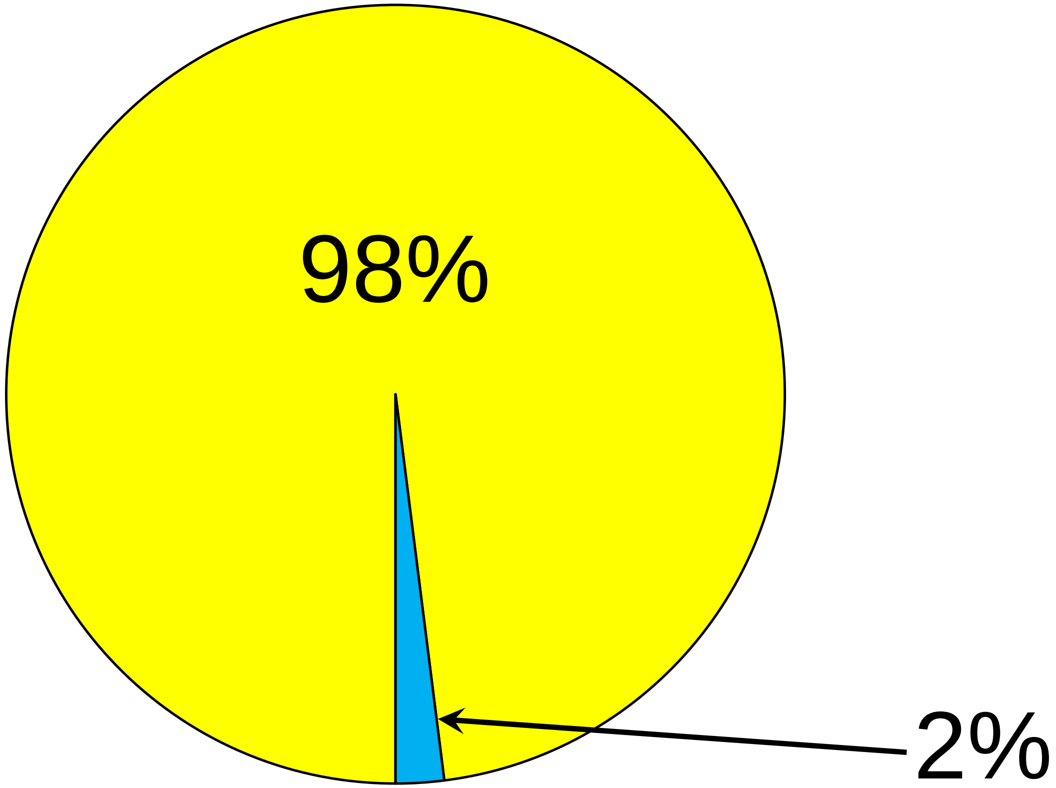
MV(E)*q <?> MV(E*q)

The value of single share 1



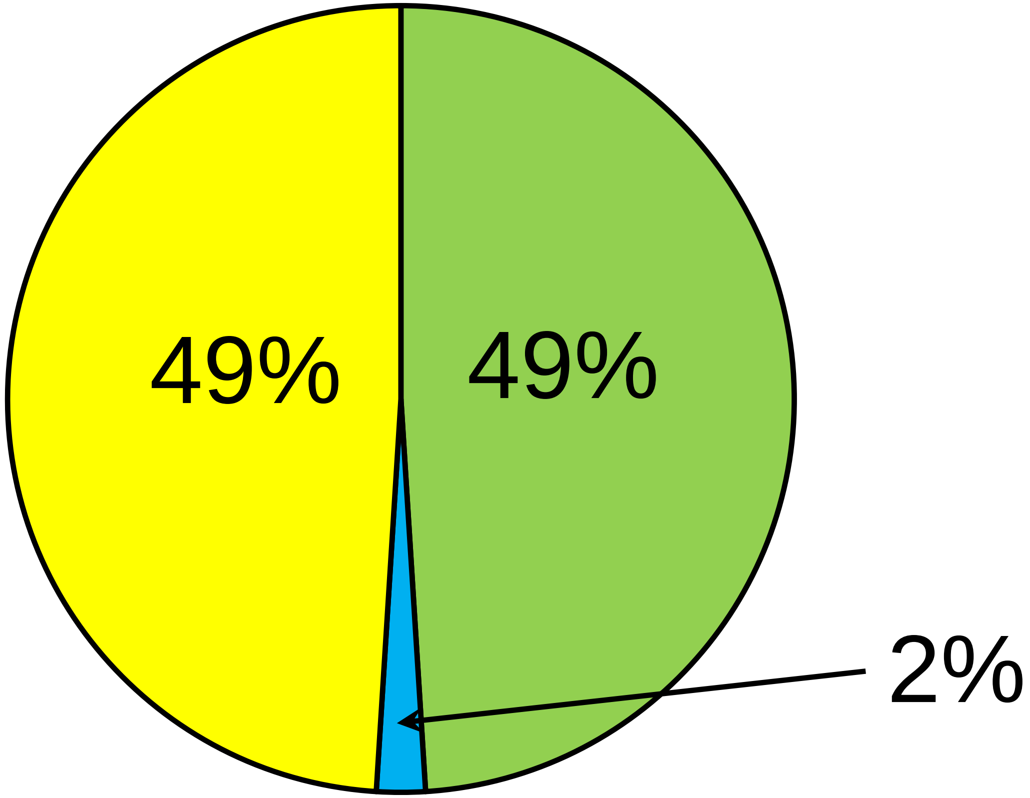
$$MV(E)*q = MV(E*q)$$

The value of single share 2



MV(E)*q >> MV(E*q)

The value of single share 3

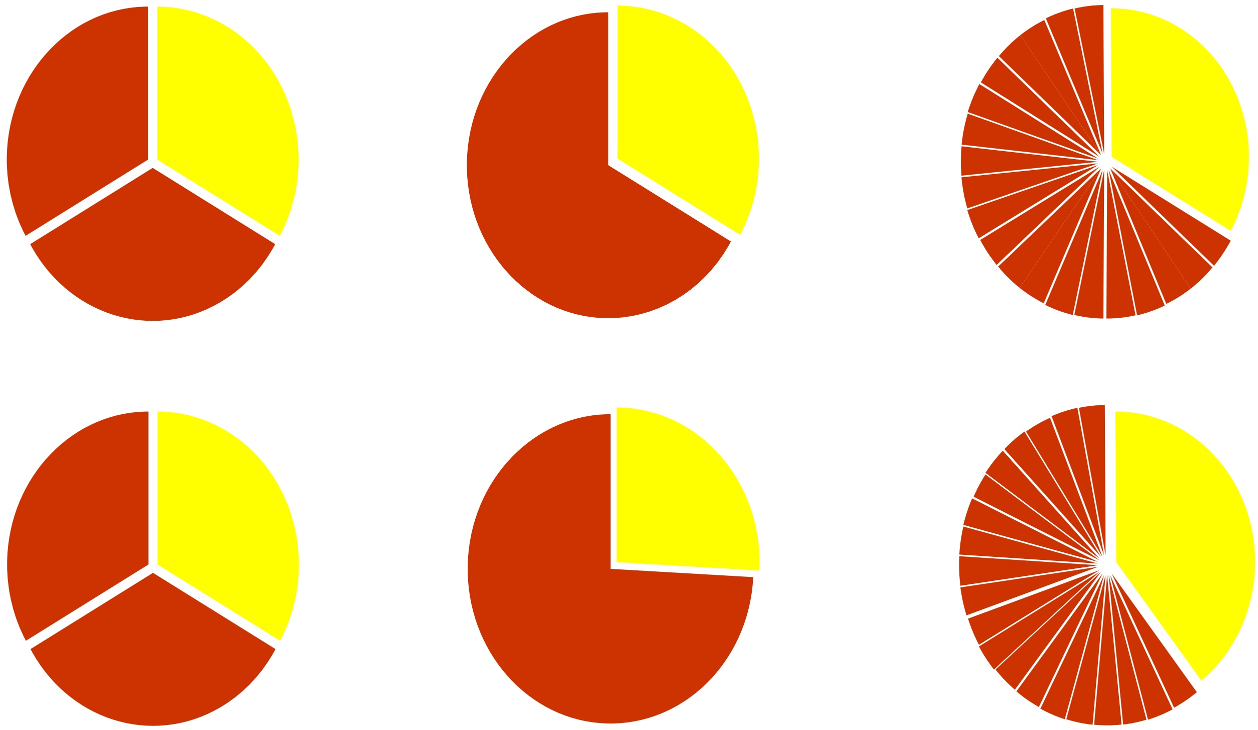


MV(E)*q << MV(E*q)

The value of a 33% stake

Ownership stake

Value stake



-30%
(or -22.1%)
discount

+30%
(or +18.2%)
premium

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The sum is always 100%

60%-40% division of the ownership rights

Control premium (%)		Minority discount (%)
10.0	60*0.10/40=	15.0
15.0	60*0.15/40=	22.5
20.0	60*0.20/40=	30.0
25.0	60*0.25/40=	37.5

II. Effect of ownership-change

What kind of opportunity to influence the operation of the firm is offered by the given stake? (Why you pay for the control is that you can make the firm more valuable using it.)

The exact size of the control premium/discount can only be estimated on a case-by-case basis.

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The value of the control

1. The likelihood of a change in the control of the firm

2. The value added by the change in the control:
   quality of the current management,
   difficulty of exercising the control

When to look for a control premium? 1

- Poor asset management
- Too conservative investment policy
- Extreme desire for investment
- Missed strategic opportunities
- M&As without synergies

When to look for a control premium? 2

- Too conservative financing structure
- Extreme indebtedness
- Poor choice of financing forms and sources
- No trust of spending the accumulated cash correctly

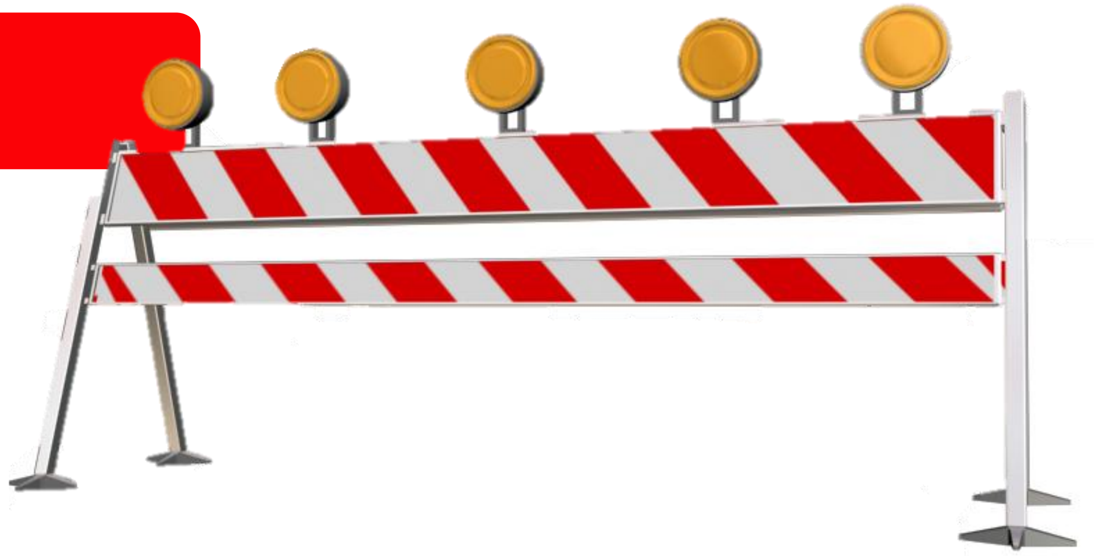
Likelihood of a change in control

Institutional barriers

Takeover regulations
Limitations of ownership
Evading conflicts of interest

Firm-specific barriers

Voting preferred shares
Cross-ownership holding structure
Significant management ownership
Very diversified ownership
Base rules limiting ownership



Value of control for listed firms

Market value = current value +

+ probability in change of control *

* (optimal value – current value)

We could do it better, but it is not for sure that we may.

Control premium for public firms

A premium may be paid if a subjective valuation dictates a higher than current market price. (Information on mistakes)

Quotes may move significantly if the estimated likelihood of a change in control would modify. (News on possible M&As)

Weaker investor protection leads to lower share prices.

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Private firms: Control premium 1

For private firms, control may have a significant value as usually management would only change if ownership changes.

Minority discount depends on the ratio of the optimal and the current equity value.

Control premium 2

Example on the difference between the valuation of stakes of 49.9%:

Current equity value= 100
Optimal equity value= 150

current value 49.9 % = 49.90
optimal value 49.9% = 74.85

control premium = $\frac{74.85}{49.90} - 1 = 50.0\%$

minority discount = $1 - \frac{49.90}{74.85} = 33.3\%$

Other stakes have the same premium!

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Control premium 3

Minority discounts are inversely linked to management abilities.

You do not always need a 50%+1 stake for the control.

The value of any stake depends heavily on whether the owner is capable and willing to be involved into the management of the firm .

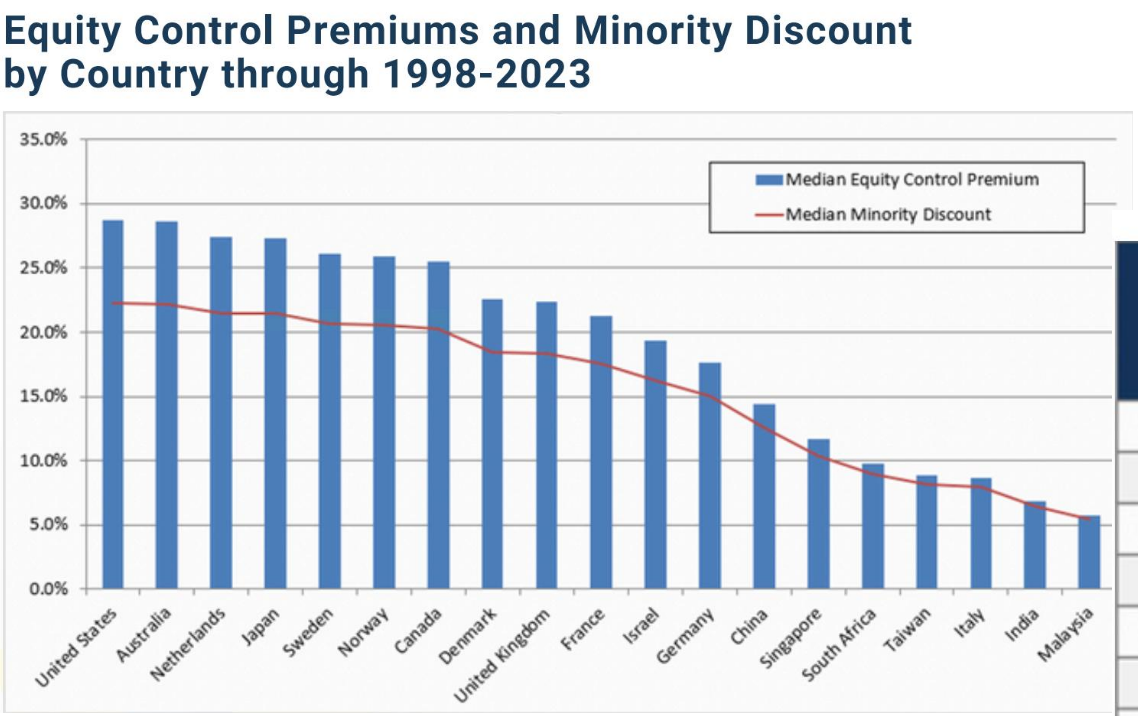
Deviations across the world 1

More efficient investment protection: 2-5% (e.g. US, UK)

Less strict: 60-65% (e.g. Brazil or Czechia)

Median: 10-12%.

Deviations across the world 2



Target Nation	Transaction Count	Average Net Sales ($ million US)	Median Net Sales ($ million US)	Median Equity Control Premium	Median Minority Discount on Equity
United States	6,713	$954	$128	28.7%	22.3%
United Kingdom	1,470	$1,145	$115	22.4%	18.3%
Canada	1,429	$320	$40	25.5%	20.3%
Japan	985	$723	$193	27.3%	21.5%
Australia	677	$373	$82	28.6%	22.2%
Hong Kong	463	$350	$40	0.0%	0.0%
France	351	$880	$106	21.3%	17.5%
India	270	$481	$66	6.8%	6.4%
Singapore	264	$395	$95	11.7%	10.4%
Sweden	255	$301	$74	26.1%	20.7%
Germany	247	$1,997	$173	17.6%	15.0%
China	229	$814	$254	14.4%	12.6%
Malaysia	217	$251	$56	5.7%	5.4%
Norway	165	$374	$119	25.9%	20.6%
Italy	154	$1,444	$203	8.7%	8.0%
Netherlands	135	$2,063	$450	27.4%	21.5%
South Africa	129	$548	$113	9.8%	9.0%
Taiwan	123	$579	$157	8.9%	8.2%
Israel	91	$247	$68	19.3%	16.2%
Denmark	82	$529	$117	22.6%	18.4%

Source: BVR, 2024 https://www.bvresources.com/docs/default-source/free-downloads/global-control-premium-stats-by-country.pdf?sfvrsn=1798f7b2_16

Everything can be sold at a low-enough price.

The level of liquidity is different across assets.

- Price effect (timing, information effect, continuity of prices)
- Alternative cost of waiting
- Transaction costs

What is liquidity? 2

High transaction fees:

real estate agents: 1-4%

action houses for paintings: 15-30%

Reasons for higher fees:
fewer agents,
non-standardised products,
more personal knowledge needed

Marketability discount

Dividend payment	Availability of information	Accessibility of potential buyers
Size of the stake	Put option	Limitations on sale

Measuring liquidity discount 1

Non-liquide investments have no well-organised markets – higher selling costs, less stable sales price

Measuring liquidity discount 2

How to quantify illiquidity discount?

- Liquidity discount in the stake or share price at the end
- Increasing required yield for liquidity risks
- Multiple-based valuation based on firms with similar liquidity

Use one method only at a time.

Setting the liquidity discount 1

Fixed discount (range) – strongly relying on former studies that have set 25-35% for shares with limited liquidity.

Many have shown that the samples were distorted and the realistic value is far smaller.

Setting the liquidity discount 2

Firm-specific discount – depends on the possible number of buyers and the difficulty of the sale. (Real estates are easier, collections are harder to sell.)

The liquidity of private firms may change from firm to firm and buyer to buyer.

Lower liquidity discount

More liquid assets	Stable, positive cash flow	Going public is near
Majority stake	Bigger firm (Costs fixed)	Operational and asset value are similar

Liquidity discount in the yields

Liquidity discount appears as a premium in required yield. **The premium desired changes over time.**

Liquidity discount may increase during times the liquidity is needed. (When market liquidity is poor asset liquidity becomes more important.)

In the long run everything is liquid. How long may you add a liquidity correction?

Liquidity discount in the yields

Returns of liquid and less liquid shares show a historical average difference of 3.0-3.5%.

Smaller firms on average had a 4.0% higher yield than S&P companies (15.7% vs 11.7%, 1984-2004).

Life cycle matters: younger companies yielded 19.9% while more mature firms offered 13.7% (vs 11.7% S&P500) (1984-2004)

Links among discounts

More liquid markets have lower lack-of-control discounts

- It is easy to sell if the majority owner does not do a good job.

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Discounts and premiums

R

Liquidity discount (assets, size, financial power, chance of going public)

Control rights (blocking minority)

Additional voting rights

Big package discount

Key person discount

Discount on portfolios without synergy

Environmental, legal, dependent liabilities discounts

Long term liabilities (contracts, warranties, guaranties, warrants)

Court decisions and market transaction, US (1980-1997)

Minority discount:

21-31%

Marketability discount:

(10) 20-60 (80) %

Big stake discount:

0-20%

Example

The HardWork Co. has three owners (40%-35%-25%). What is the value of the 35% stake, if the DCF value of the total equity amounts to 100 under the current strategy? (All parties agree on the valuation.)

If investors only focus on their own intention to buy…

Outside buyer

Own stake after the deal:		
	100 * 35% =	35.0
Private firm (total of equity)		
(liquidity, transparency)		-10%
Strategic minority		+ 5%
	35 * 0.90 * 1.05 =	33.1
Premium used	33.1 / 35.0 – 1 =	-5.4%

25% owner

Own stake until now:	100 * 25% =	25.0
Illiquidity of the stake:		-10%
	25 * 0.90 =	22.5
Own stake after the deal:		
	100 * 0.60 =	60.0
Control premium		+17%
	60 * 1.166 =	70.0
Value of 35% stake	70 – 22.5 =	47.5
Premium used	47.5 / 35 – 1 =	35.7%

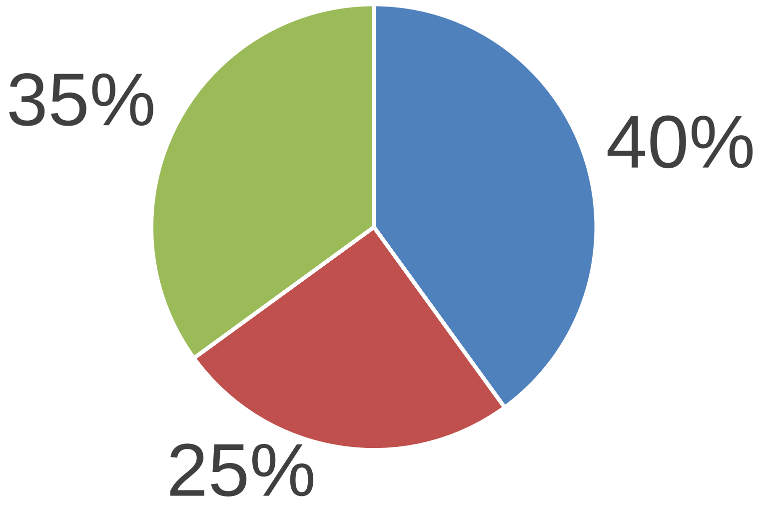
40% owner

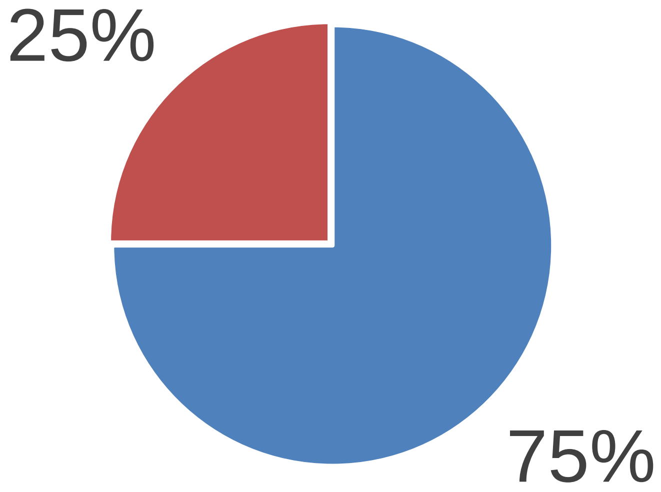
Own stake until now:	100 * 40% =	40.0
Own stake after the deal:		
	100 * 0.75 =	75.0
Strategic control premium		+13%
	60 * 1.1333 =	85.0
Value of 35% stake	85.0 – 40.0 =	45.0
Premium used	45 / 35 – 1 =	28.6%

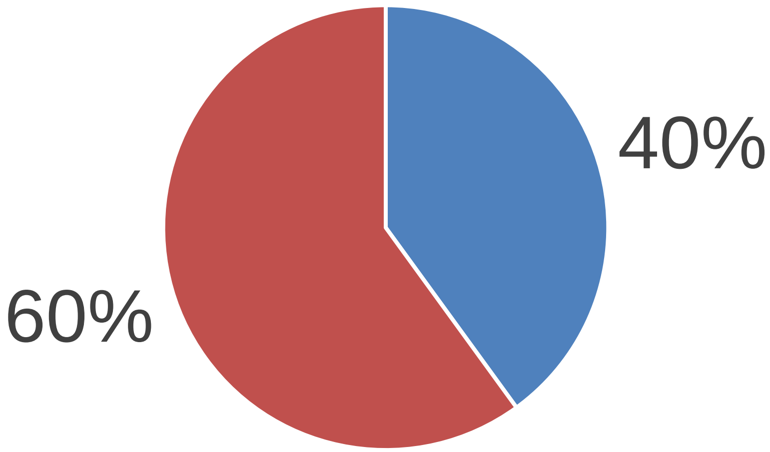
What if everyone is interested?

Owners do not contrast to the current status rather than to **the future status they will be in if not purchasing**.

Possible outcomes







For the 25% owner

Own stake if 40% owner buys:		25.0
Liquidity and minority discount		-40%
	25 * 0.60 =	15.0
Own stake (25%+35%) after the deal:		
	100 * 0.60 =	60.0
Control premium		+17%
	60 * 1.166 =	70.0
Value of 35% stake	70.0 – 15.0 =	55.0
Premium used	55 / 35 – 1 =	57.1%

For the 40% owner

Own stake if 25% owner buys :		40.0
Liquidity and minority discount		-25%
	40 * 0.75 =	30.0
Own stake (40%+35%) after the deal :		
	100 * 0.75 =	75.0
Control premium		+13%
	60 * 1.20 =	85.0
Value of 35% stake	85.0 – 30.0 =	55.0
Premium used	55 / 35 – 1 =	57.1%

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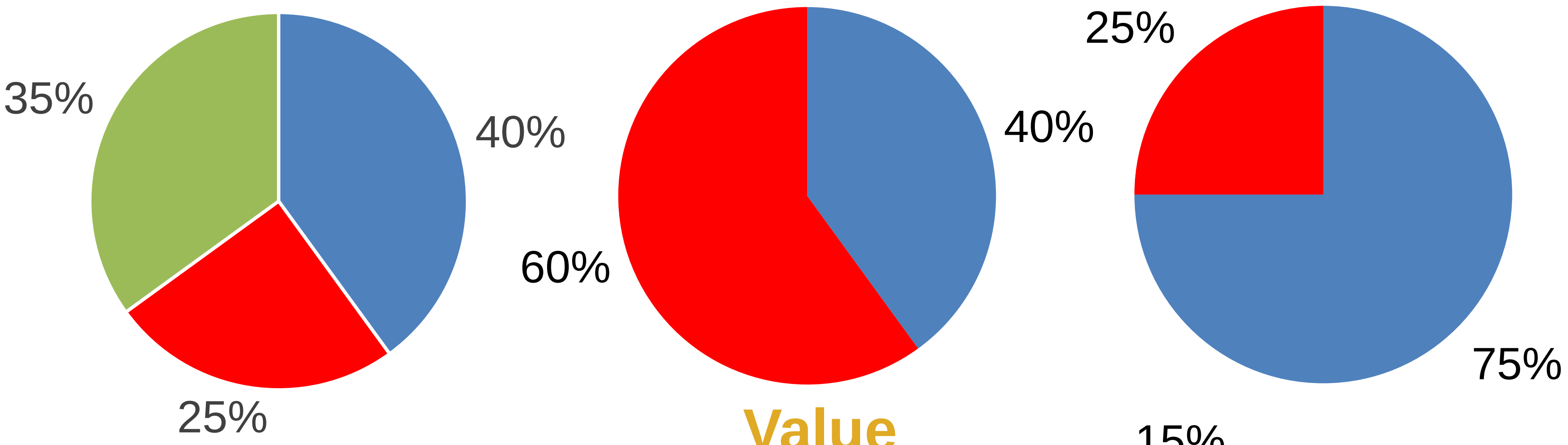
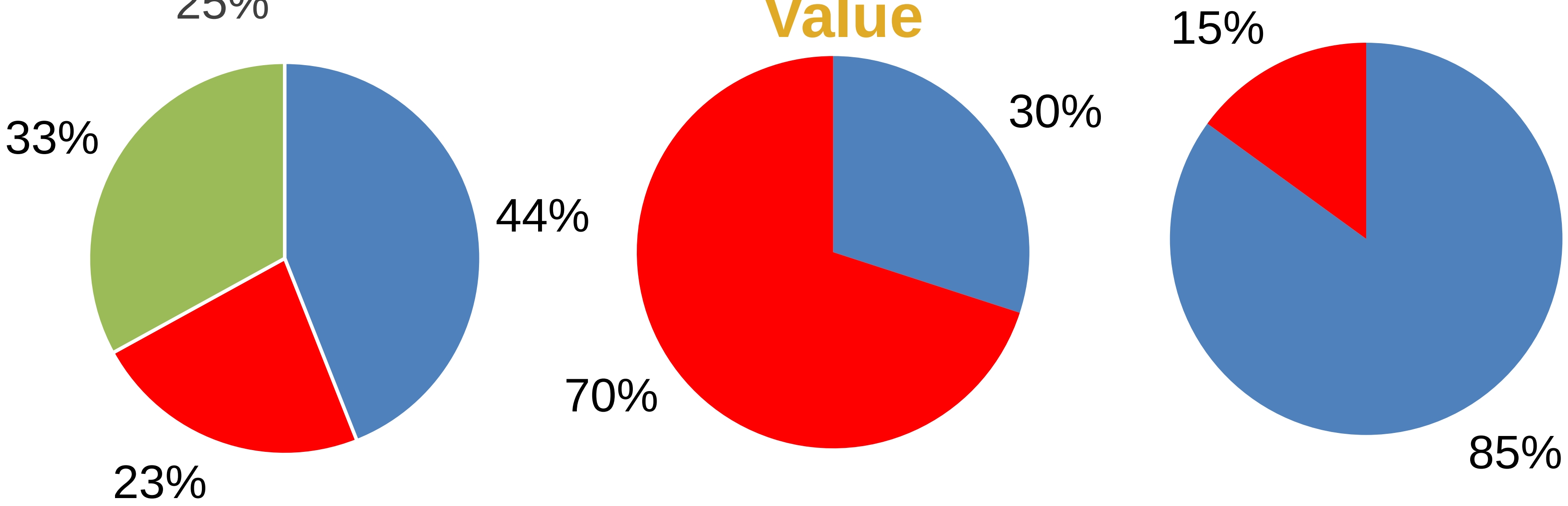
Dividing value across stakes

Ownership



Value



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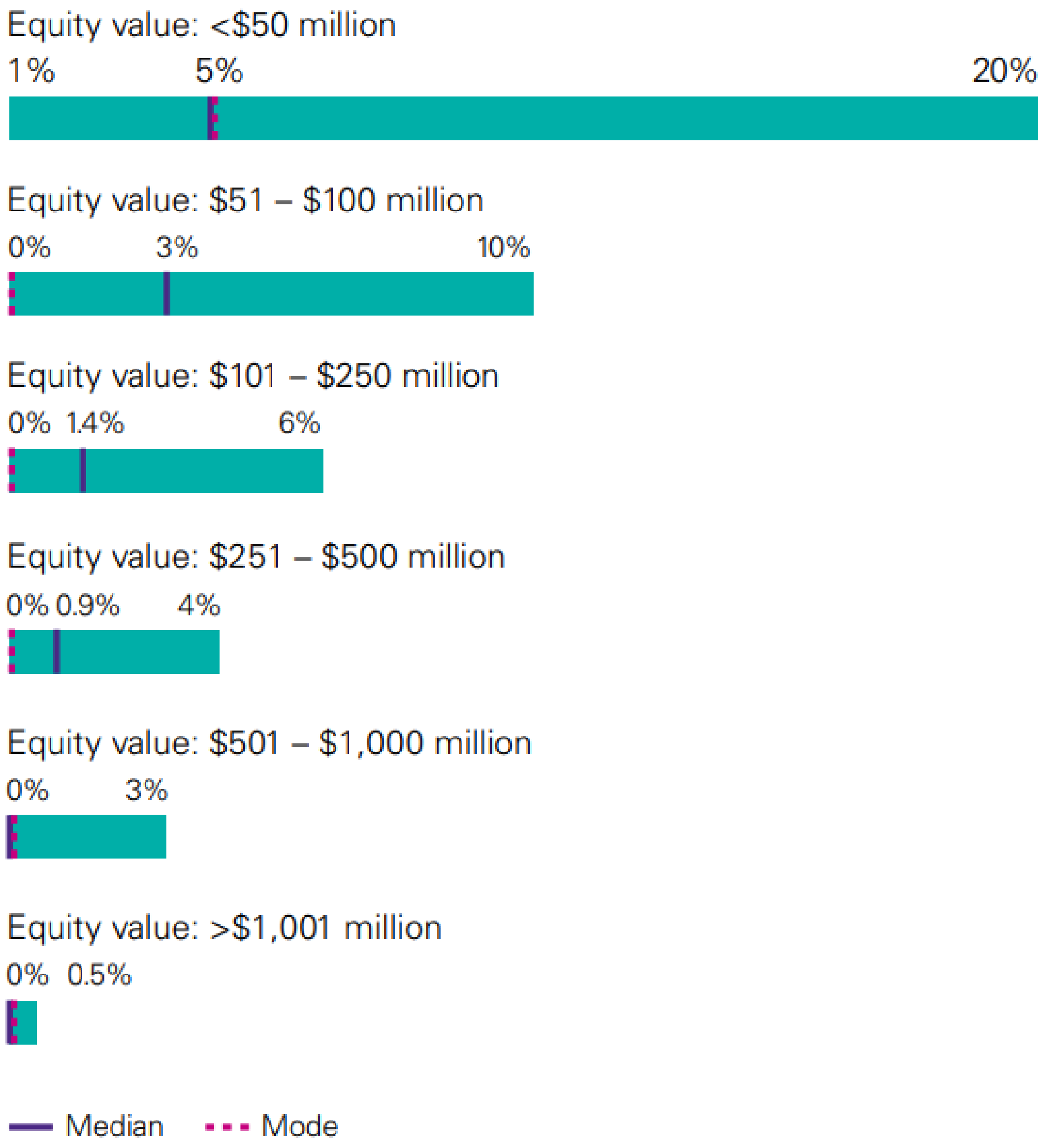
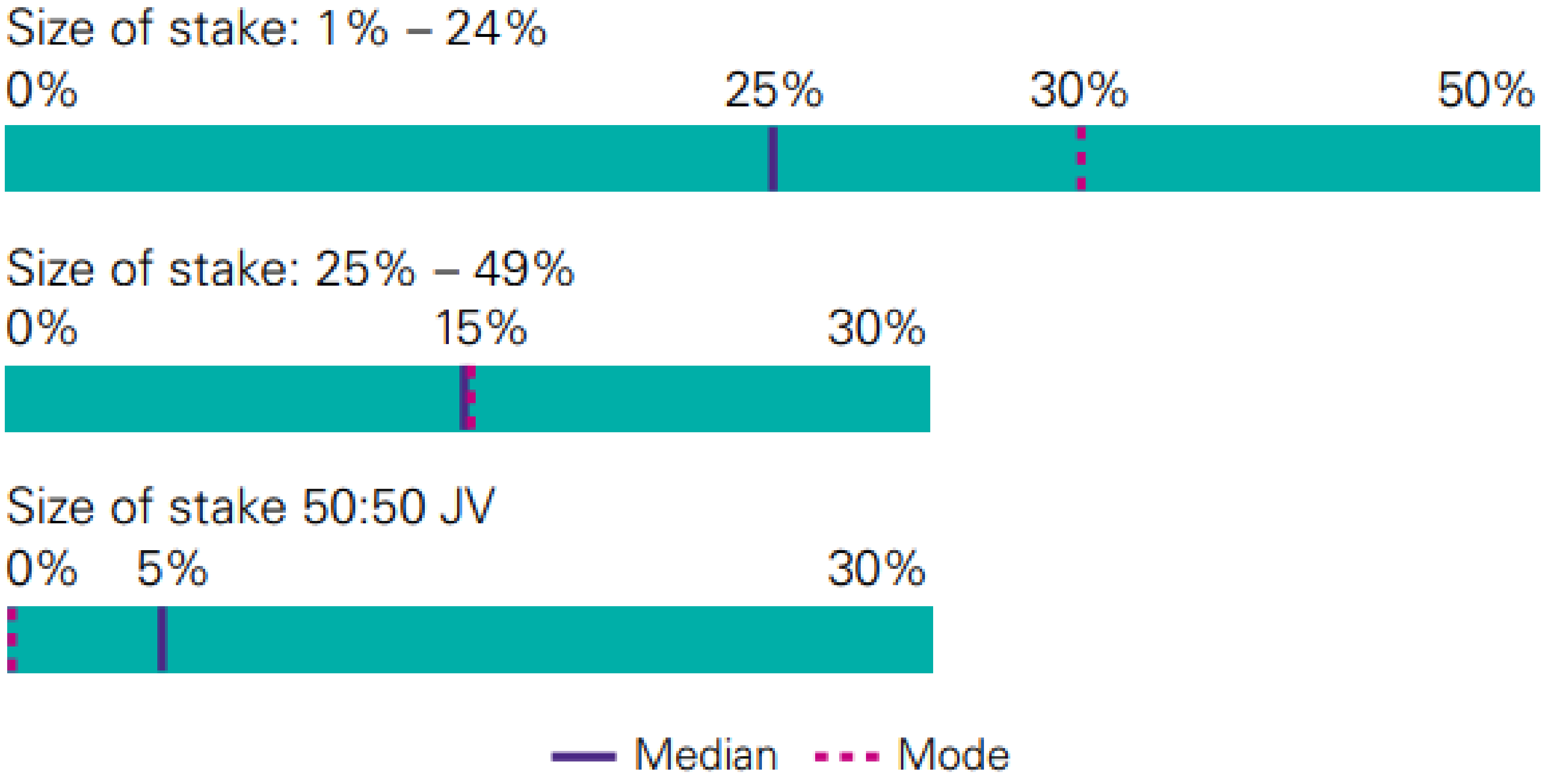
How is it done in the industry?

What premium is added to the discount rate to account for the size of the entity being valued?



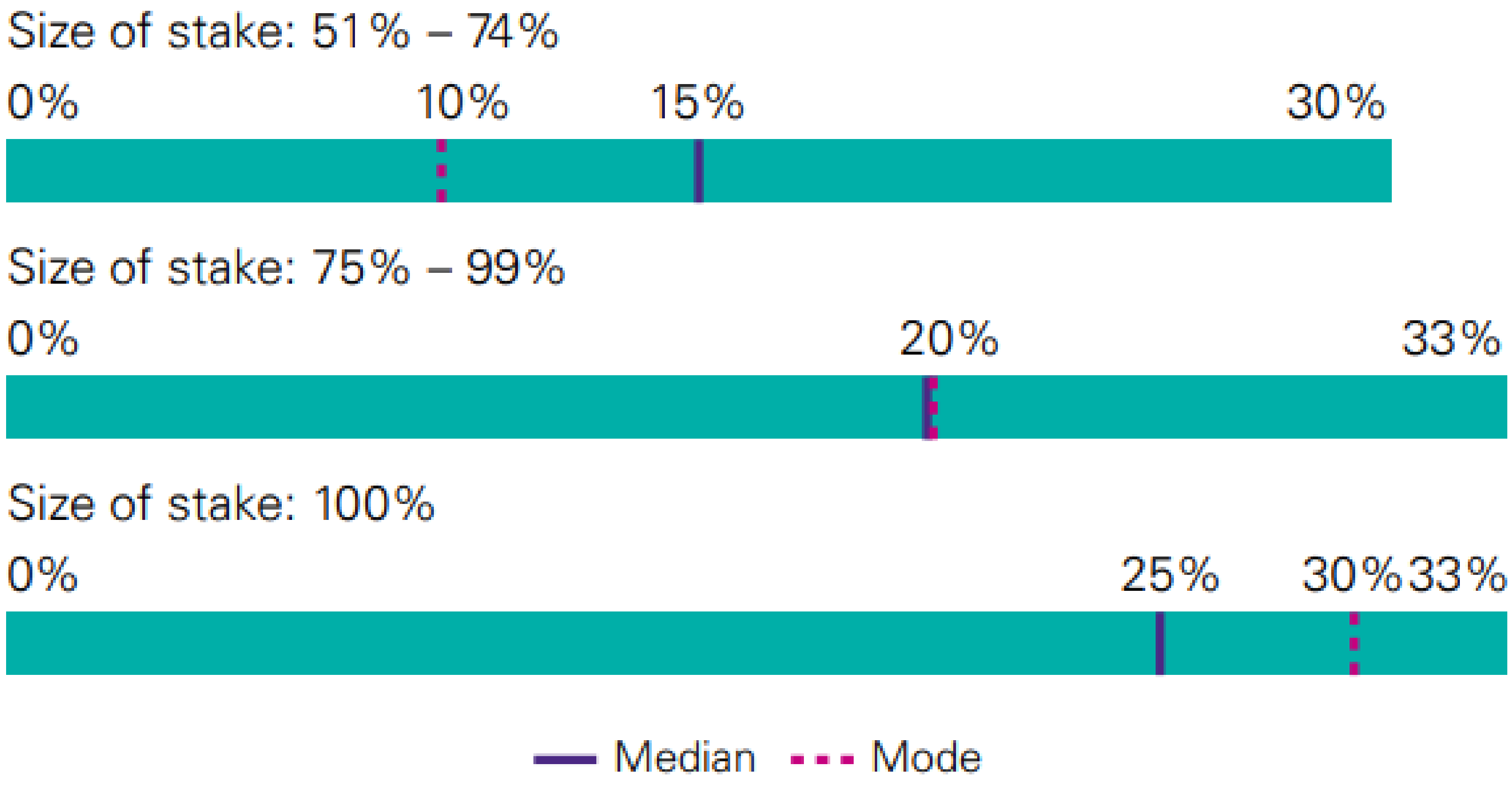
Source: For all it's worth – KPMG Valuation Practices Survey 2017 Australia https://assets.kpmg/content/dam/kpmg/au/pdf/2017/valuation-practices-survey-2017.pdf

What discount do you apply in valuing a minority interest?



Source: For all it’s worth –
KPMG Valuation Practices Survey 2017 Australia

What premium do you apply to reflect the level of control offered by the size of the stake being valued?



Source: For all it's worth –
KPMG Valuation
Practices Survey 2017
Australia

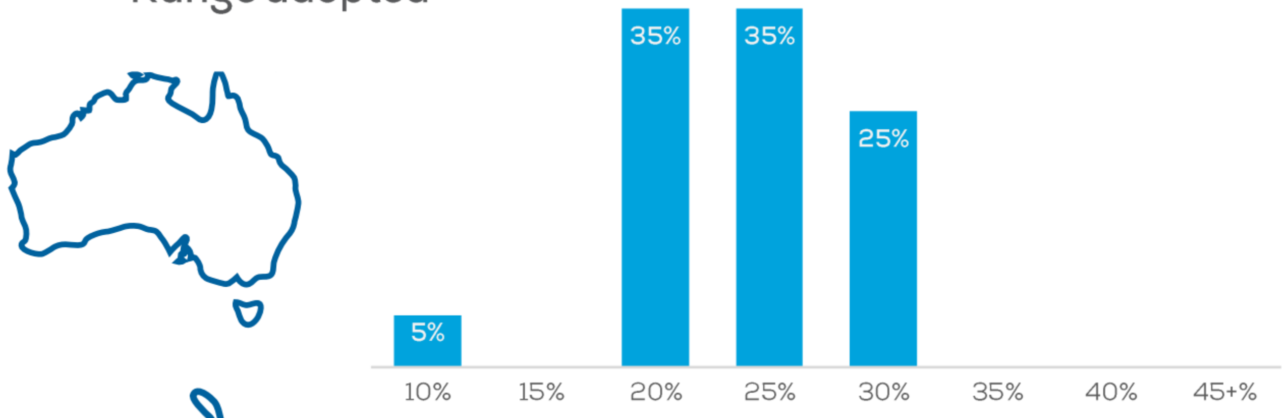
For those using a standard control premium, the most common range adopted is 20 - 25%

Use of standard control premium

57%

43%

Range adopted



New Zealand – 20%

Malaysia – 10%

Notes: sample size n = 10.1: 51, 10.2: 22
Q10.1: Do you use a standard control premium?
Q10.2: What range do you currently adopt as the standard control premium in the {Country} market?

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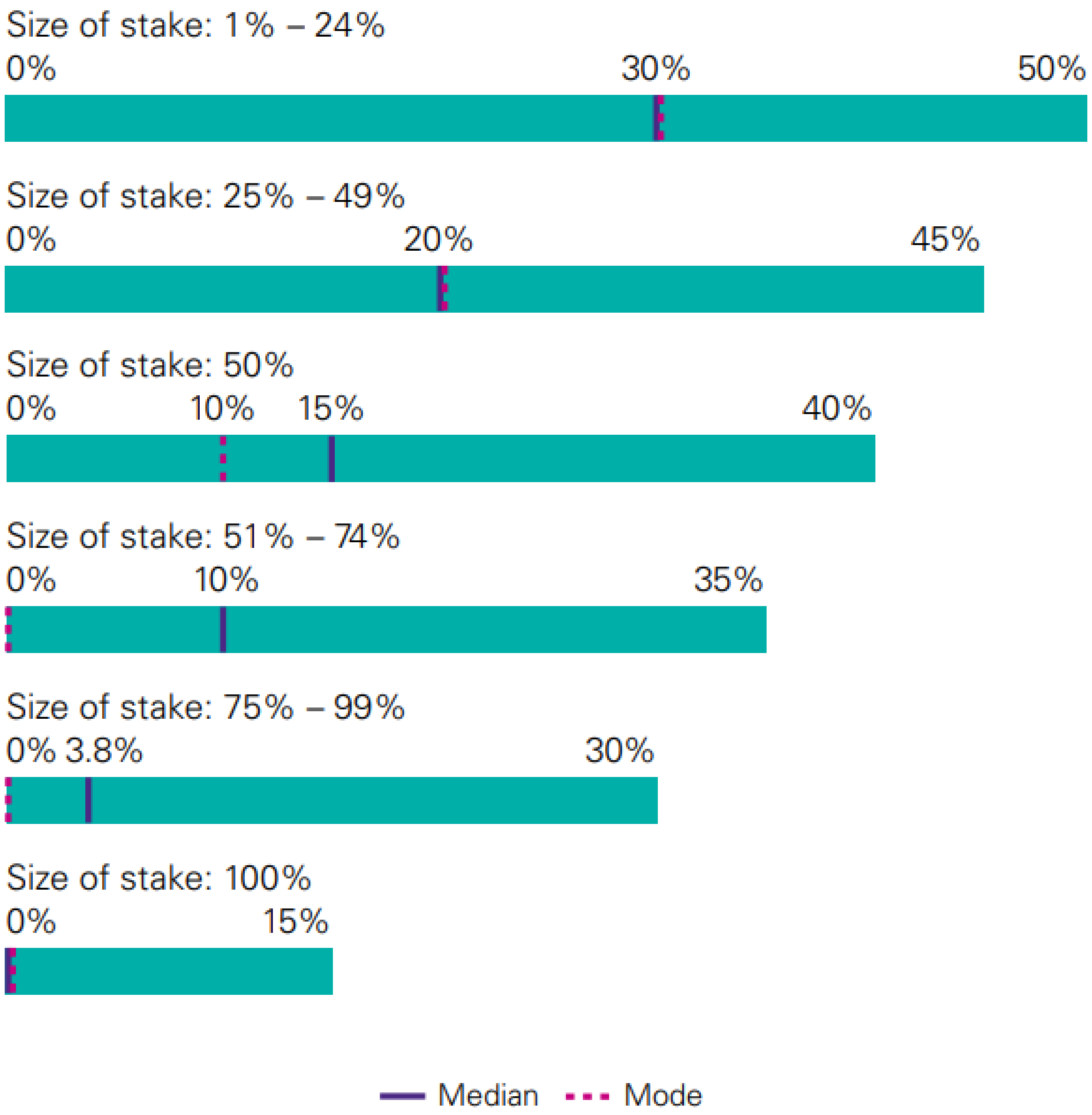
When a marketability discount is required, what sized discount do you apply?



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Thank you for your attention!